25901444 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEPHY205 - THEORY OF RELATIVITY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **3** marks

1. From velocity composition law, show that speed of light is the same in two different inertial frames. [U] / [CO1]
2. Draw the spacetime diagram of a particle moving with uniform velocity. [U] / [CO1]
3. Does the speed of light have same value for two observers related by Galilean transformation? [U] / [CO1]
4. Show that the outer product of (2,0) and (0,2) type tensor is a (2,2) tensor. [A] / [CO2]
5. Given $A^{\mu} = (A^0, A^1, A^2, A^3)$, write explicitly the components of A_{μ}. [An] / [CO2]
6. Write the inverse transformation corresponding to a rotation by angle θ. [A] / [CO2]
7. How does the Minkowski metric classify vectors as timelike, spacelike, or null? [U] / [CO2]
8. Express plane polar basis vectors $(\vec{e}_r, \vec{e}_\theta)$ in terms of Cartesian bases $(\vec{e}_x, \vec{e}_y)$. [U] / [CO3]
9. Explain what are tidal forces. [U] / [CO3]

10. Express polar unit basis vectors $(\vec{e}_{\hat{r}}, \vec{e}_{\hat{\theta}})$ in terms of polar basis vectors $(\vec{e}_r, \vec{e}_\theta)$. [U] / [CO3]

11. Write the expression for covariant derivative of a contravariant vector component $V^{\mu}{}_{;\nu}$. [U] / [CO4]

12. Explain Geometrized units. [U] / [CO4]

[3x10 = 30]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Inertial frame R finds an event to occur at time $t = 3\ \text{meters}$ at coordinates $x = 2\ \text{meters}$, $y = 0$, $z = 0$. Find the invariant hyperbola for this event in $c = 1$ units. [A] / [CO1]

14. Show that when relative velocity between two frames, $v \ll c$ Lorentz transformation reduces to Galilean transformation. [U] / [CO1]

15. Observer R carries a lamp that he flashes 100 times every minute. What is the frequency of light flashes for observer R' who is moving uniformly at $0.6c$ with respect to R? [A] / [CO1]

16. Obtain the nonrelativistic limit for the equation $E^2 = p^c c^2 + m^2 c^4$. [An] / [CO2]

17. For a lightlike four-vector k^μ, What is the value of $k^\mu k_\mu$? [An] / [CO2]

18. Find the number of independent components of a symmetric tensor of type (0,2). Is it the same for (2,0) type tensor? Explain [An] / [CO2]

19. Prove the symmetry $\Gamma^{\mu}{}_{\rho\sigma} = \Gamma^{\mu}{}_{\sigma\rho}$ of Christoffel symbol components. [U] / [CO3]

20. FInd the derivatives of Cartesian basis vectors $(\vec{e}_x, \vec{e}_y)$ with respect to plane polar coordinates (r, θ). [U] / [CO3]

21. Define parallel transport and explan its significance. [U] / [CO4]

22. How many independent components $R_{\mu\nu\rho\sigma}$ does Riemann tensor have if $(\mu, \nu, \rho, \sigma = 0, 1, 2)$? [A] / [CO4]

[5x8 = 40]